

CLIMATIC AI

Leveraging India's National Data for Climate Intelligence

Team VOYAGER-1

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MISSION OVERVIEW

India's Climate, Seen From Orbit

Climatic AI fuses ISRO satellite data with ground sensor networks to deliver real-time, predictive climate intelligence — built for a country of 1.4 billion people facing rising heat, erratic monsoons, and coastal risk.

50+

Weather stations feeding
live rainfall data

24/7

Continuous satellite &
sensor monitoring

3

Oceans tracked for climate
signals



Why It Matters

Fragmented data slows disaster response. Climatic AI unifies satellite imagery, atmospheric sensors, and ocean buoys into one predictive engine — turning raw signals into early warnings.

Four Pillars of Climate Tech

1

Perovskite Solar

Record-efficiency panels capturing more energy at lower manufacturing cost.

2

Green Hydrogen

Clean electrolysis fuel for heavy industry, shipping, and aviation.

3

Direct Air Capture

Removing CO2 at the source, enabling permanent carbon storage.

4

AI Smart Grids

IoT-optimized distribution that integrates renewables and cuts waste.

Environmental Metrics We Track



Temperature

Real-time urban & rural tracking



Air Quality

PM2.5, PM10, CO2, Ozone



Rainfall

50+ weather stations, seasonal patterns



Tree Cover

Satellite forest & green-space monitoring



Humidity

68% current · 72% monthly avg



Biodiversity

142 species · index 3.8 (healthy)

ISRO API Integration

INSAT & Resourcesat satellite constellations feed our AI prediction engine directly.

1

Fetching Atmospheric Data

Temperature gradients, humidity layers, and pressure systems streamed for climate modeling.

2

Processing Satellite Imagery

AI pipeline detects cloud formation, land change, and oceanic heat patterns.

3

Real-Time Dashboard

Continuous visualization of anomalies, storm trajectories, and instant alerts.

Live Satellite Monitoring

Real-time satellite tracking capabilities powering early detection across India.



Live Satellite Map

Continuously updated orbital positioning and coverage.



ISRO Satellite Feed

Direct high-resolution imagery stream.



Rainfall Tracking

Precipitation patterns across regional grids.



Cyclone Monitoring

Storm trajectory and intensity forecasting.

Three Oceans Under Study

Pacific Ocean

El Niño & La Niña cycles, coral reef health, carbon absorption rates.

Atlantic Ocean

Gulf Stream velocity, hurricane dynamics, and AMOC circulation.

Indian Ocean

Monsoon dynamics, coral bleaching, and marine heatwave detection.

Detection & Early Warning

Advanced monitoring that turns raw satellite signal into life-saving alerts.



Earthquake Detection

Seismic pattern recognition for rapid regional alerts.



Heatwave Alert

Tracks rising temperature anomalies to protect vulnerable regions.



Pollution Warning

Monitors air quality & water contamination thresholds daily.



Wildlife Scanner

Satellite-based habitat and biodiversity conservation tracking.

Meet Nebula

A voice-powered AI assistant delivering intelligent climate insights across environmental industries.



Voice Recognition

Natural speech commands for hands-free climate insights.



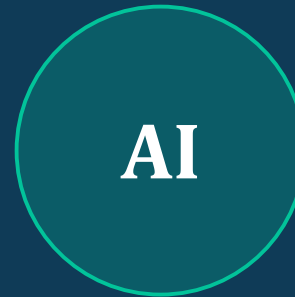
Climate Data Analysis

Processes vast atmospheric datasets to surface patterns instantly.



Smart Recommendations

AI-driven action guidance tailored to location and need.



www.climatictech.com

THANK YOU

Climatic AI — Harnessing Space Technology for Climate Intelligence

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